K-Field Spectrophotometer Spherical Coherence

Five-Revolution Overlapping Reconstruction and ICRS Unit-Sphere Analysis

P. Dwayne Esterline, Principal Investigator

BS, Huntington University (1999); MBA, Indiana Wesleyan University (2008)

Krytronx, LLC.

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Core result: elementwise summation of five consecutive, phase-aligned device revolutions at one-revolution cadence increases unit-sphere coherence in every measured channel while preserving the same dominant degree-2, even-parity spherical geometry.

This report is a self-contained educational and reconstruction reference. Every numerical result and figure is generated from the pic_id14A experimental record and its exact one-turn, three-turn, and five-turn unit-sphere analysis ledgers.

Abstract

This archival reference presents the complete five-revolution overlapping reconstruction of the pic_id14A rotating spectrophotometer experiment. The source contains 46,710 spectra organized as 519 complete revolutions with 90 indexed positions per revolution. For every spectral channel, five consecutive revolutions were aligned by device index and summed elementwise. The window advanced by one revolution, producing 515 overlapping stacked profiles and 46,350 sphere-registered values per channel. Each stack was registered to the effective UTC and inertial direction of its center revolution. The resulting trajectories were placed on an ICRS unit sphere and accumulated in a 72 by 36 equal-area longitude-sin(latitude) grid containing 2,592 cells. The trajectory directly sampled 1,344 cells, or 51.8519 percent of the spherical area.

The five-turn reconstruction increased the sphere eta^2 coherence statistic in every channel relative to both the single-turn/raw mapping and the three-turn reconstruction. The strongest five-turn results were 480 nm with eta^2 = 0.969747, split-map correlation = 0.993206, and spherical-harmonic cell R^2 = 0.968012; 445 nm with eta^2 = 0.933965; and NIR with eta^2 = 0.909296. The lower-coherence channels also improved systematically: 415 nm reached 0.757237, 555 nm reached 0.600390, Clear reached 0.585296, 680 nm reached 0.438892, 590 nm reached 0.407008, 515 nm reached 0.349337, and 630 nm reached 0.316982. Every channel remained dominated by spherical-harmonic degree l = 2. The even-degree power fraction ranged from 0.890896 to 0.999862.

The measured improvement establishes that the spherical structure is not confined to unresolved single-revolution samples. Multi-turn phase-aligned integration raises the signal above the channel noise floor while preserving the same broad quadrupolar and antipodal geometry. The method changes the analysis unit from an isolated detector sample to a coherently reconstructed multi-revolution state, providing a general pathway for low-SNR rotating experiments in which the signal requires several overlapping revolutions to resolve.

Keywords: K-Field; rotating spectrophotometer; ICRS; unit sphere; overlapping revolutions; coherent summation; spherical harmonics; quadrupole; antipodal coherence.

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1. Scientific Scope and Central Result

The scientific question is whether the full timewise response of each spectrophotometer channel forms a coherent structure when its measured detector orientations are placed in a common inertial unit-sphere coordinate system. A single revolution does not fully resolve the signal of interest. The measured structure emerges over approximately three to ten overlapping revolutions, while individual revolutions remain strongly affected by the detector and photon-count noise floor. Therefore, the correct analysis does not treat each revolution as an independent final signal. It first constructs a phase-aligned multi-turn signal and only then maps that reconstructed signal into inertial space.

The five-turn analysis uses the same device phase at every term of the sum. Position index 0 from five consecutive revolutions is summed with position index 0; position index 1 is summed with position index 1; and so forth through position index 89. The operation increases coherent signal amplitude in direct proportion to the number of aligned revolutions. For independent zero-mean noise, noise amplitude grows only as the square root of the number of revolutions. Five-turn summation therefore has an ideal coherent-amplitude signal-to-noise gain of sqrt(5) relative to a single revolution and sqrt(5/3) relative to a three-turn sum.

The complete result is not merely an increase in amplitude. Coherence increases in every channel while the dominant spherical geometry remains unchanged: all ten channels retain degree l = 2 as their dominant real spherical-harmonic degree, and all retain strong even-degree power. Thus, the multi-turn operation strengthens a persistent spatial structure rather than creating a new structure with a different topology.

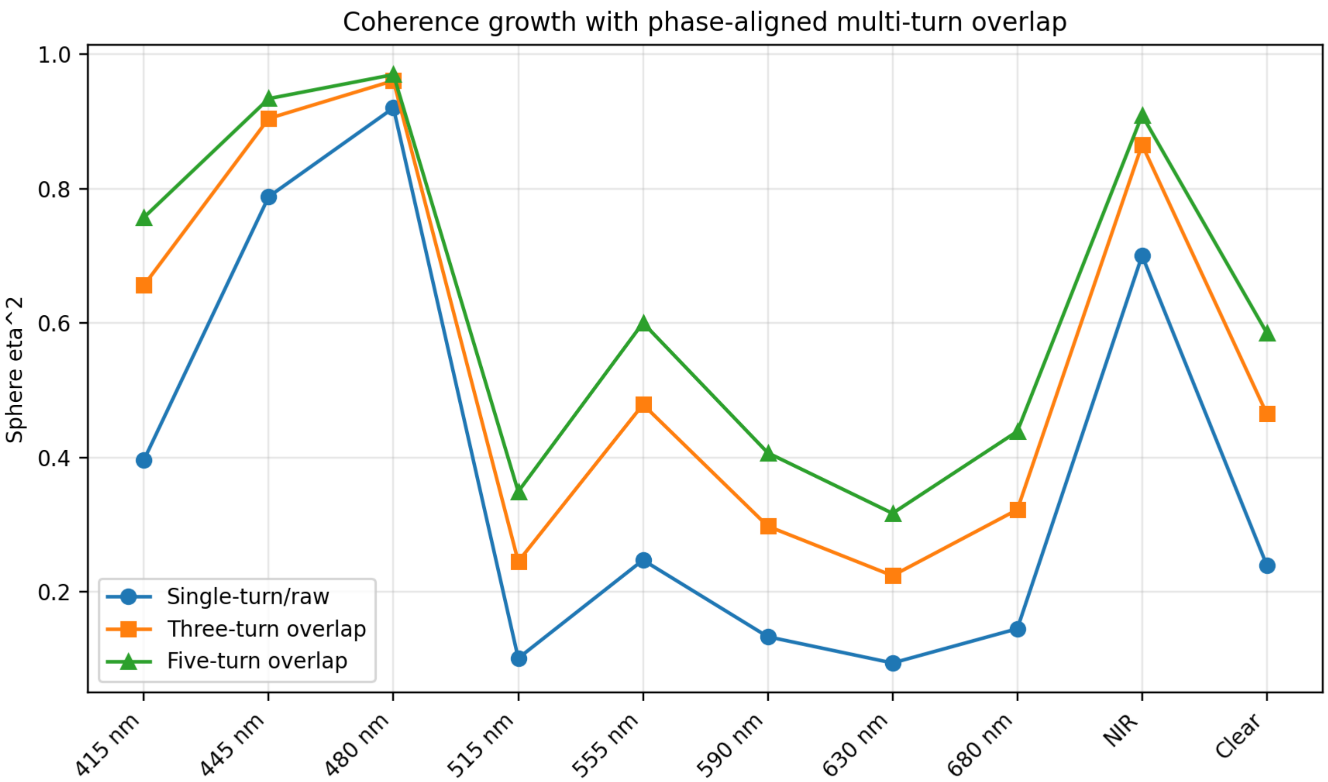

Figure 1. Sphere eta^2 for the single-turn/raw, three-turn, and five-turn representations. Every channel rises from three to five turns. The strongest channels approach eta^2 = 1, while weaker channels show the largest relative gains.

2. Experimental Record

2.1 Instrument channels

The experiment used a rotating multichannel spectrophotometer record with ten simultaneously measured outputs: nominal 415, 445, 480, 515, 555, 590, 630, and 680 nm spectral channels, an NIR channel, and a broadband Clear channel. The detector completed one indexed revolution through a vertical north-up-south-down scan plane for every 90 measurements. The device index therefore advances in 4 degree increments.

2.2 Source size and temporal coverage

Quantity	Exact value
Source rows	46,710 spectra
Complete revolutions	519
Positions per revolution	90
First recorded UTC	2026-08-04 10:55:12.918393
Last recorded UTC	2026-08-04 17:32:58.329756
Total recorded duration	6.629280934 h
Earth rotation during record	99.711469427 deg
Mean revolution period	45.984421956 s
Median revolution period	45.984914000 s
Revolution-period standard deviation	0.013339648 s
Mean indexed sample interval	0.510938022 s

The mean revolution period is approximately 45.9844 s, so a five-turn reconstruction uses five phase-matched observations separated by one revolution. The first and fifth observations at a given device index are separated by approximately 183.9377 s. The full five-revolution temporal support is approximately 229.9221 s.

2.3 Effective time

Each sample uses its effective acquisition time rather than the later table-write time. The effective time is reconstructed as the recorded UTC timestamp minus the frame-read interval. This correction places the orientation basis at the time represented by the measurement.

```
t_effective = t_date_UTC - t_frame_read
```

3. Device-to-ICRS Coordinate Transformation

3.1 Local scan geometry

The detector axis is horizontal and east-west. The rotating sensing direction therefore moves through the local north-up-south-down vertical plane. Index zero points downward. With theta = 4 deg times device index, the unit direction in inertial coordinates is constructed from the time-dependent local north and up basis vectors.

```
u(t,theta) = sin(theta) north_ICRS(t) - cos(theta) up_ICRS(t)
```

At theta = 0 deg, u = -up and the detector points down. At theta = 90 deg, u = north. At theta = 180 deg, u = up. At theta = 270 deg, u = -north and the detector points south. Earth rotation changes the north and up vectors continuously, so consecutive laboratory circles occupy different inertial great-circle traces.

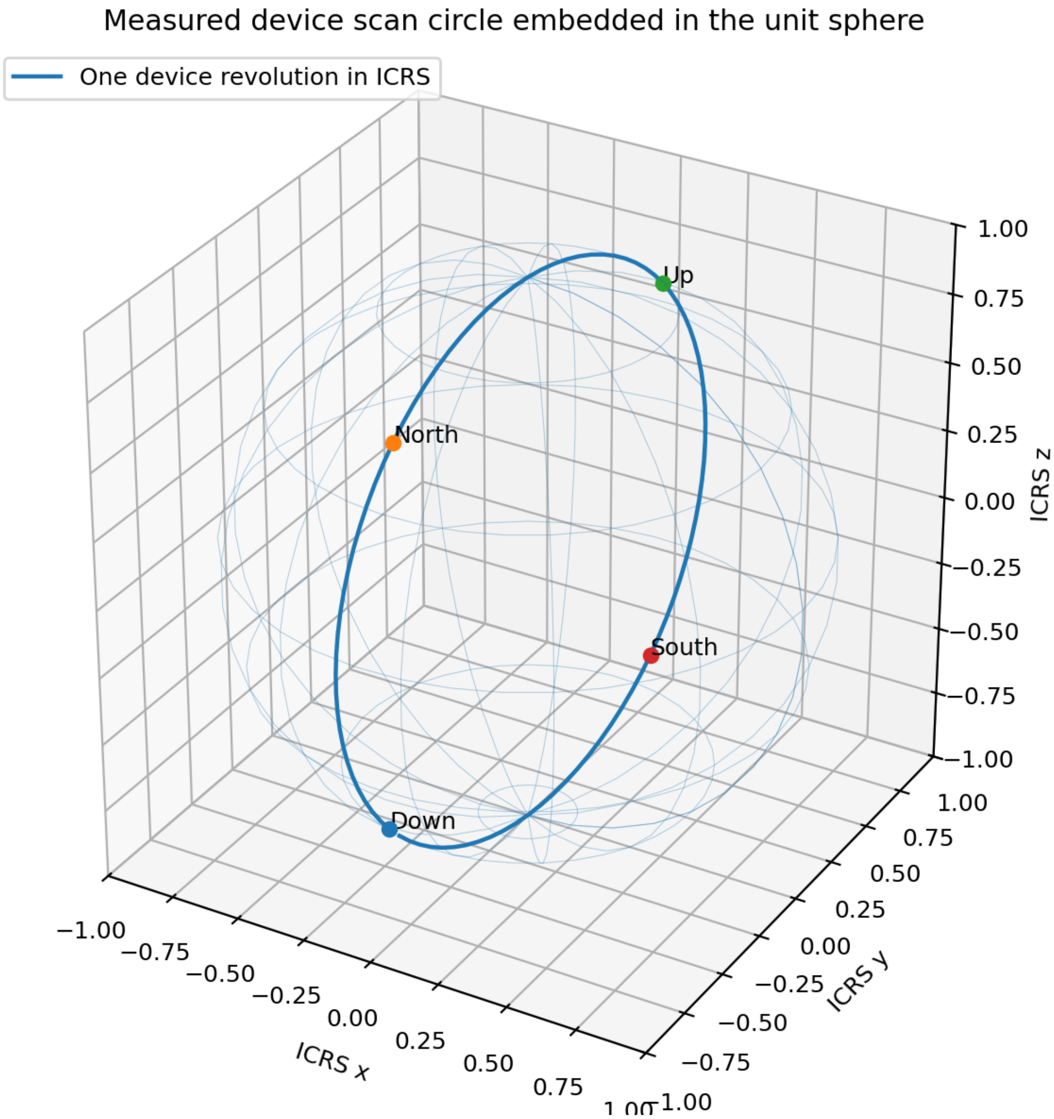

Figure 2. Dimensionally correct unit-sphere embedding of an actual 90-position center revolution. The plotted points are the measured ICRS direction coordinates assigned to one reconstructed five-turn profile. The four cardinal device orientations are labeled.

3.2 Angular smearing from center registration

Each five-turn stack is registered to its center revolution. The maximum temporal offset between the center and either outer phase-matched observation is two mean revolutions, or 91.968844 s. Earth rotates by 0.192126346 deg per device revolution, so the maximum center-to-edge inertial displacement is 0.384252693 deg and the complete first-to-fifth displacement is 0.768505386 deg. This is the angular-resolution cost paid for the five-sample coherent sum. It is substantially smaller than the 4 deg mechanical index spacing and smaller than the approximately 4 deg equivalent angular size of one equal-area surface cell.

Smearing quantity	Value
Earth rotation per device revolution	0.192126346 deg
Center to first/last contributing revolution	0.384252693 deg
First to fifth contributing revolution	0.768505386 deg
Mechanical angular sample spacing	4.000000000 deg
Equivalent equal-area cell side	3.989422804 deg

4. Five-Revolution Overlapping Reconstruction

4.1 Elementwise overlay

Let x_c,k(theta_i) denote channel c at device index i during revolution k. The five-turn reconstructed signal is the elementwise sum

$$S_{c,k}(\theta_i) = x_{c,k}(\theta_i) + x_{c,k+1}(\theta_i) + x_{c,k+2}(\theta_i) + x_{c,k+3}(\theta_i) + x_{c,k+4}(\theta_i)$$

The cadence is one revolution, so the next stack begins at revolution k + 1. Adjacent stacks share four of their five revolutions. This overlap is intentional: it preserves timewise continuity and permits the resolved multi-turn signal to move across the inertial sphere at the natural Earth-rotation cadence.

4.2 Stack count and registration

Item	Value
Input complete revolutions	519
Window length	5 consecutive revolutions
Cadence	1 revolution
Number of overlapping stacks	515
Positions per stack	90
Sphere-mapped values per channel	46,350
Registration epoch	Center revolution, k + 2
Detrending	None

The first two and final two revolutions cannot serve as centers of a symmetric five-turn window, leaving 515 valid center revolutions. Every stack retains 90 device positions. No 901-sample moving trend is subtracted because that operation overlaps the signal-resolution timescale and would suppress the multi-turn structure being measured.

4.3 Signal-to-noise interpretation

For a phase-stable component with amplitude A per revolution, the five-turn sum has amplitude 5A. For independent noise with standard deviation sigma per revolution, the sum has standard deviation sqrt(5) sigma. The ideal amplitude SNR becomes sqrt(5) A/sigma. The operation does not assume the observed channel noise is perfectly independent; it provides the correct coherent integration direction, and the measured coherence statistics determine the realized improvement.

5. Equal-Area Unit-Sphere Grid

5.1 Grid definition

The inertial sphere is partitioned into 72 longitude bins and 36 equal-sin(latitude) bands. Longitude bins are exactly 5 deg wide. Latitude boundaries are uniform in sin(latitude), not in latitude itself. This makes every cell have the same solid angle.

$$\Delta\Omega = \Delta \text{longitude} \times \Delta \sin(\text{latitude}) = (2\pi/72) \times (2/36) = 4\pi/2592$$

Grid property	Value
Longitude bins	72
Equal-sin(latitude) bands	36
Total cells	2,592
Solid angle per cell	0.004848136811 sr
Equivalent square angular side	3.989422804 deg
Directly sampled cells	1,344
Directly sampled area fraction	0.518518519

Grid property	Value
Median samples per occupied cell	26

5.2 Dimensionally correct base cell

Cell 1313 is an actually occupied representative cell near the equator. Its longitude bounds are 85.0 to 90.0 deg and its latitude bounds are 0.000000000 to 3.184738537 deg. The four corner nodes are listed below in ICRS unit-vector coordinates. Edge lengths are angular arc lengths on a sphere of radius one.

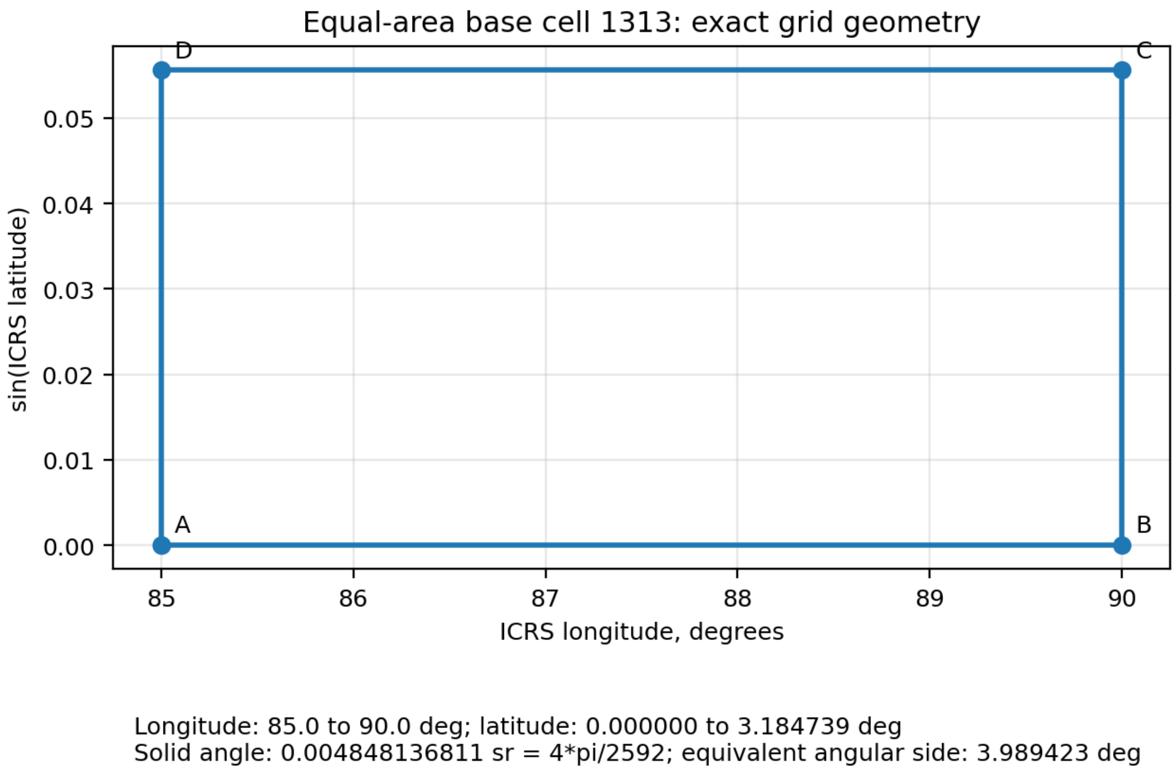

Figure 3. Exact equal-area base cell 1313 in the longitude-sin(latitude) coordinate plane. The rectangular shape in this plane is the defining reason each cell has equal solid angle. Nodes A-D are the four exact spherical corner nodes.

Node	Longitude (deg)	Latitude (deg)	ICRS x	ICRS y	ICRS z
A	85.000000000	0.000000000	+0.087155742748	+0.996194698092	+0.000000000000
B	90.000000000	0.000000000	+0.000000000000	+1.000000000000	+0.000000000000
C	90.000000000	3.184738537	+0.000000000000	+0.998455597534	+0.055555555556
D	85.000000000	3.184738537	+0.087021139204	+0.994656172543	+0.055555555556

Base-cell edge	Unit-sphere arc length	Angular value
A-B, lower parallel	0.087266462600 rad	5.000000000 deg
D-C, upper parallel	0.087131688060 rad	4.992277988 deg
A-D and B-C, meridians	0.055584173281 rad	3.184738537 deg

5.3 Actual coverage

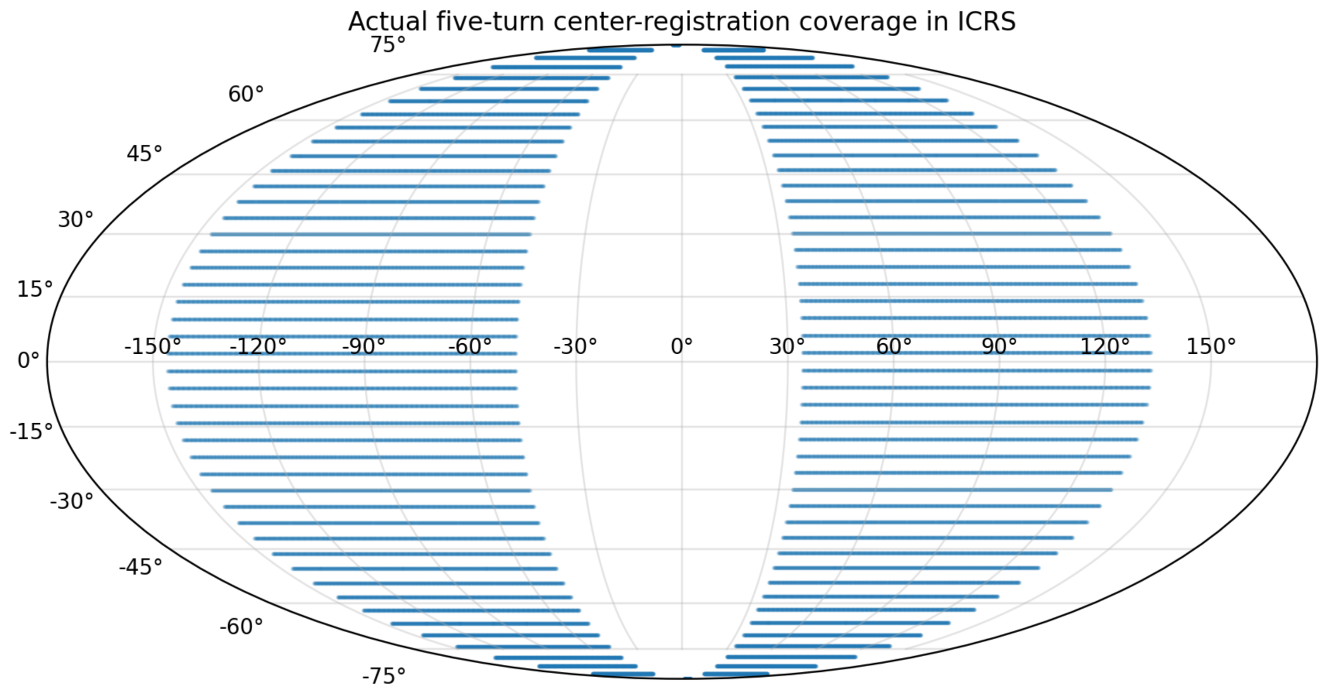

Figure 4. All 46,350 center-registered five-turn direction samples on an ICRS Mollweide projection. The measured path covers 1,344 of 2,592 equal-area cells while Earth rotates by nearly 100 deg during the experiment.

6. Coherence Measures

6.1 Sphere eta^2

Sphere eta^2 measures the fraction of total sample variance associated with the equal-area sphere-cell means. Let y_j be a stacked signal value, g(j) its sphere cell, y_bar the global mean, and y_bar_g the mean in cell g. Then

```
eta^2 = sum_g n_g (y_bar_g - y_bar)^2 / sum_j (y_j - y_bar)^2
```

A value near zero means that sphere location explains little of the measured variation. A value near one means that samples assigned to the same inertial cells are highly consistent relative to the total variation.

6.2 Split-map correlation

The stacked time series is divided into alternating ten-stack blocks. A separate weighted cell-mean map is constructed for each half, and the maps are correlated over common cells. This tests whether the same broad surface pattern appears in separated portions of the record. Because adjacent five-turn stacks share four contributing revolutions, the effective number of independent map realizations is smaller than the raw stack count.

6.3 Whole-turn circular-shift null

The null family circularly shifts each stacked channel by whole revolution units. Device phase and temporal autocorrelation are preserved while ICRS registry is changed. Sixty-four shifts are evaluated. In an overlapping five-turn sequence, nearby shifted realizations remain correlated with the original because the reconstructed windows share source revolutions. Consequently, the null percentile is conservative but not an independent-trial probability.

6.4 Spherical-harmonic representation

Real spherical harmonics through degree l = 12 are fitted to occupied cell means with a weak squared-Laplacian ridge penalty. The harmonic cell R^2 reports the weighted fraction of cell-mean variance explained by the smooth spherical representation. Degree l = 1 is dipolar, degree l = 2 is quadrupolar, and higher degrees represent progressively finer angular structure. Even degrees are unchanged by the antipodal transformation u -> -u, so a large even-degree power fraction indicates strong antipodal symmetry.

7. Five-Turn Results

7.1 Complete channel table

Channel	eta^2	eta null pct	Split corr.	Split null pct	Harmonic R^2	Dominant l	Even power
415 nm	0.757237	100.0000	0.912163	54.6875	0.907301	2	0.994310
445 nm	0.933965	100.0000	0.977260	73.4375	0.951382	2	0.999497
480 nm	0.969747	100.0000	0.993206	87.5000	0.968012	2	0.999862
515 nm	0.349337	85.9375	0.598008	25.0000	0.426777	2	0.986563
555 nm	0.600390	100.0000	0.833279	62.5000	0.814444	2	0.940709
590 nm	0.407008	100.0000	0.612596	6.2500	0.487020	2	0.979200
630 nm	0.316982	56.2500	0.504538	12.5000	0.378243	2	0.960627
680 nm	0.438892	84.3750	0.718098	40.6250	0.664832	2	0.898787
NIR	0.909296	100.0000	0.977249	81.2500	0.962122	2	0.998939
Clear	0.585296	100.0000	0.789715	32.8125	0.745580	2	0.982644

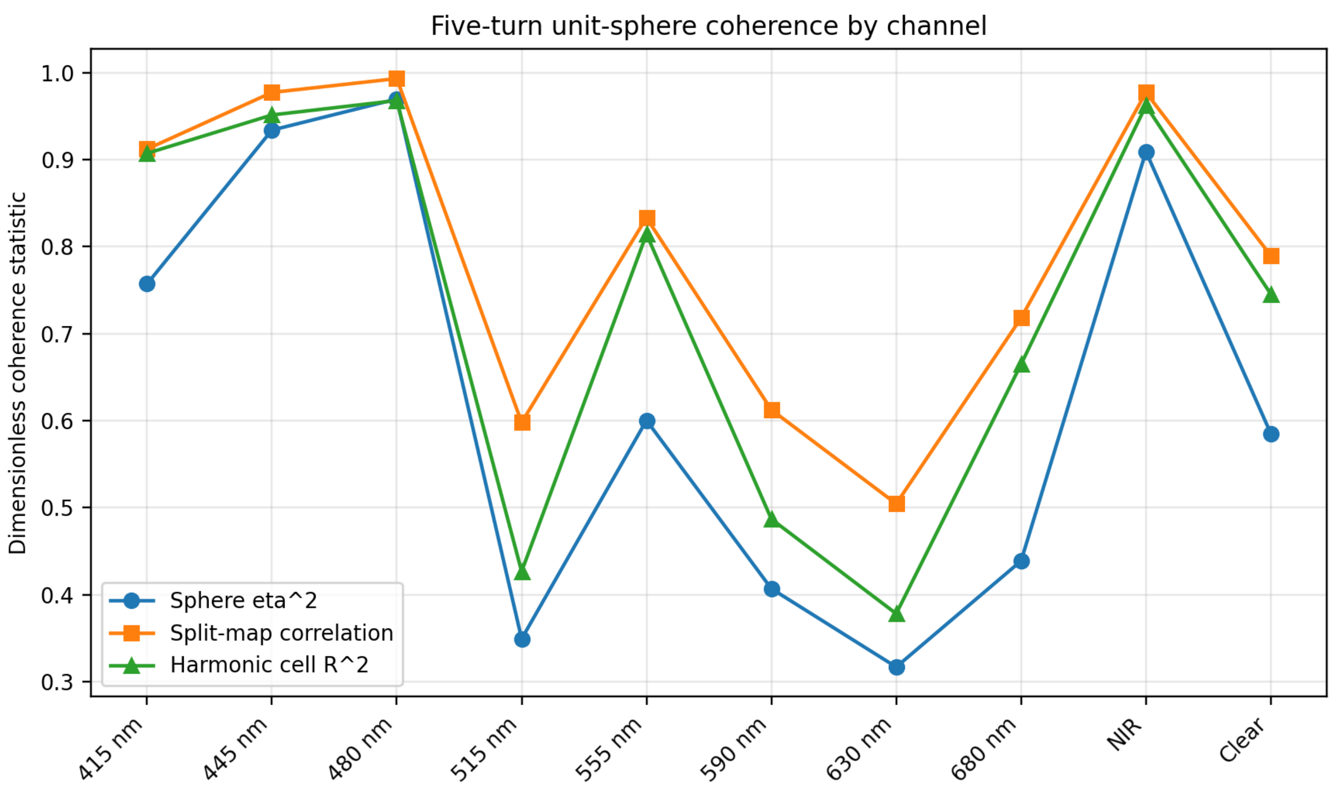

Figure 5. The three principal five-turn coherence statistics. The 480 nm, 445 nm, and NIR channels form the strongest group. The 415 nm, 555 nm, and Clear channels form a second coherent group. The remaining channels retain positive sphere structure with lower total explanatory strength.

7.2 Systematic increase from three to five turns

Channel	Single/raw eta^2	3-turn eta^2	5-turn eta^2	5-3 gain	5/raw ratio
415 nm	0.396506	0.656222	0.757237	+0.101015	1.910
445 nm	0.788082	0.904446	0.933965	+0.029519	1.185
480 nm	0.920165	0.960656	0.969747	+0.009091	1.054
515 nm	0.101413	0.245332	0.349337	+0.104005	3.445
555 nm	0.247466	0.478892	0.600390	+0.121498	2.426
590 nm	0.133405	0.297836	0.407008	+0.109173	3.051
630 nm	0.094315	0.224225	0.316982	+0.092758	3.361
680 nm	0.145318	0.322415	0.438892	+0.116478	3.020
NIR	0.700506	0.865384	0.909296	+0.043912	1.298
Clear	0.239664	0.464965	0.585296	+0.120331	2.442

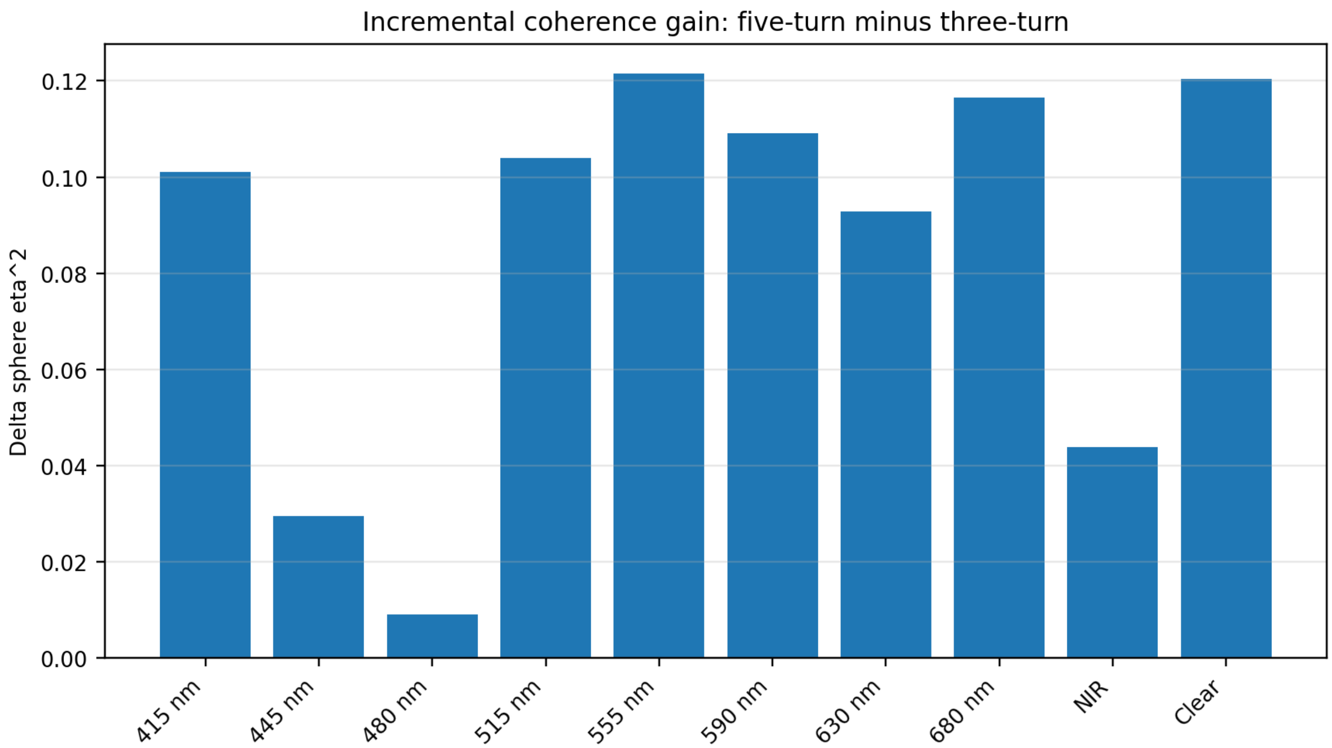

Figure 6. Incremental sphere-coherence gain from the three-turn to the five-turn reconstruction. Every bar is positive.

The five-revolution elementwise overlay increases sphere coherence in every channel. The measured transitions are 415 nm: 0.656222 to 0.757237; 445 nm: 0.904446 to 0.933965; 480 nm: 0.960656 to 0.969747; 515 nm: 0.245332 to 0.349337; 555 nm: 0.478892 to 0.600390; 590 nm: 0.297836 to 0.407008; 630 nm: 0.224225 to 0.316982; 680 nm: 0.322415 to 0.438892; NIR: 0.865384 to 0.909296; and Clear: 0.464965 to 0.585296.

The largest absolute three-to-five gains occur in Clear (+0.120331), 555 nm (+0.121498), 680 nm (+0.116477), 590 nm (+0.109172), 515 nm (+0.104005), and 415 nm (+0.101015). These channels benefit most from the additional coherent integration. The strongest channels already approach the upper bound and therefore show smaller absolute gains.

7.3 Strongest channels

The 480 nm channel is the strongest five-turn surface: eta^2 = 0.969747, split-map correlation = 0.993206, and harmonic cell R^2 = 0.968012. The 445 nm channel follows with eta^2 = 0.933965, split-map correlation = 0.977260, and harmonic cell R^2 = 0.951382. NIR reaches eta^2 = 0.909296, split-map correlation = 0.977249, and harmonic cell R^2 = 0.962122. These three channels combine high within-cell consistency, high temporal repeatability, and a highly compressible smooth spherical map.

7.4 Intermediate channels

The 415 nm channel reaches eta^2 = 0.757237 with harmonic R^2 = 0.907301. The 555 nm channel reaches eta^2 = 0.600390 with harmonic R^2 = 0.814444. Clear reaches eta^2 = 0.585296 with harmonic R^2 = 0.745580. These channels show strong recovery under five-turn integration and retain broad low-degree geometry.

7.5 Lower-amplitude but coherent channels

The 680 nm, 590 nm, 515 nm, and 630 nm channels have five-turn eta^2 values of 0.438892, 0.407008, 0.349337, and 0.316982. Their improvement is systematic, but their harmonic maps explain less cell variance. The five-turn operation therefore resolves additional sphere organization without making all spectral channels equivalent. Channel-dependent coupling strength remains a central measured feature.

8. Measured Five-Turn Surface Maps

Each map below shows the actual occupied equal-area cell means for one channel. Longitude is ICRS longitude from 0 to 360 deg; latitude is ICRS latitude from -90 to +90 deg. The map values are five-revolution elementwise sums in native detector-count units. Empty cells are not interpolated into the measured layer. The smooth harmonic fit is retained separately in the machine-readable companion.

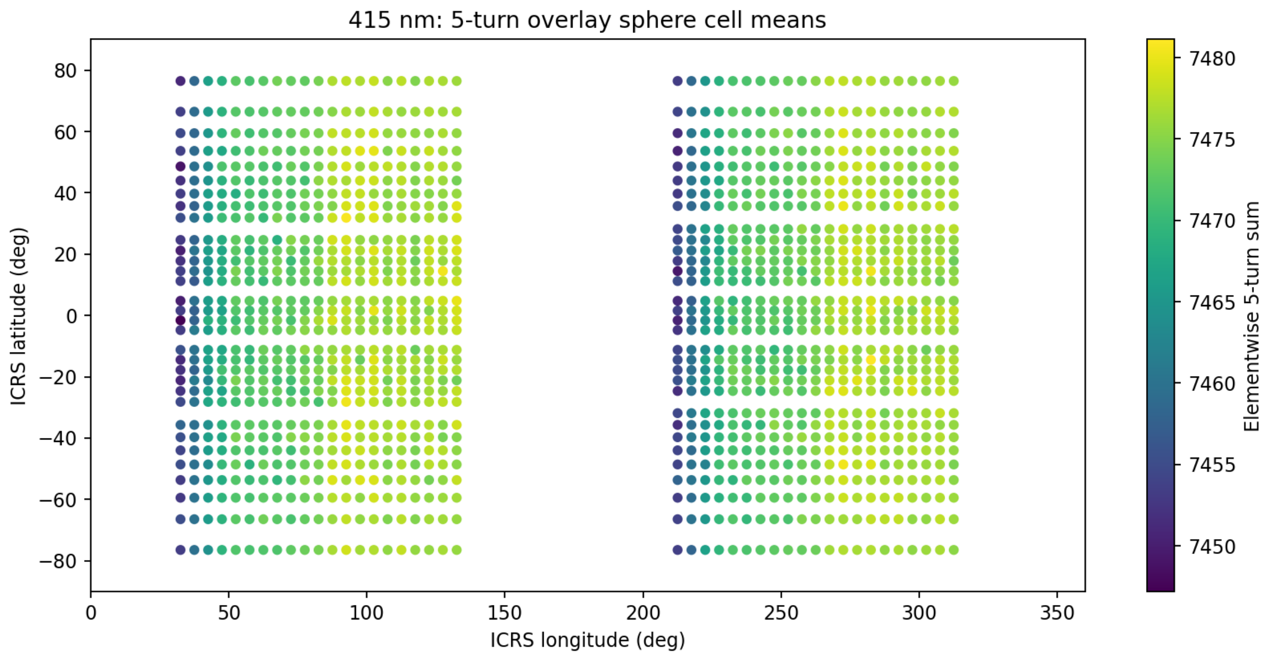

Figure 7. 415 nm five-turn surface cell means. This figure is generated directly from the 515 overlapping five-turn profiles and their center-registered ICRS coordinates.

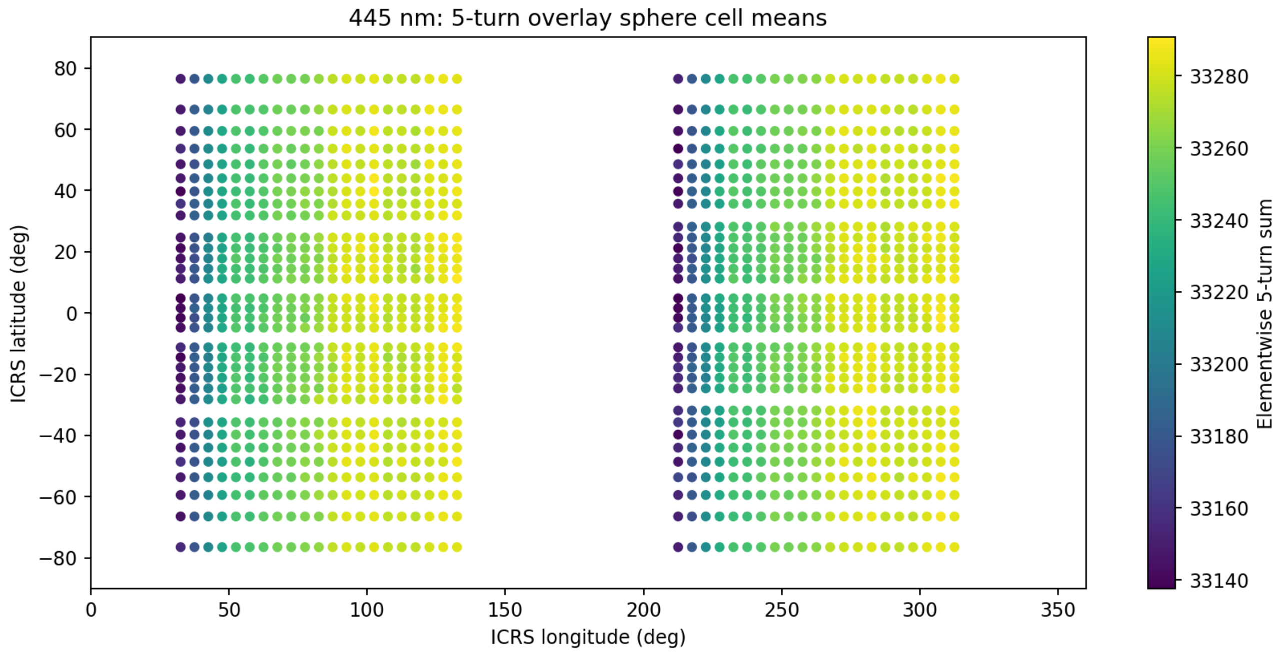

Figure 8. 445 nm five-turn surface cell means. This figure is generated directly from the 515 overlapping five-turn profiles and their center-registered ICRS coordinates.

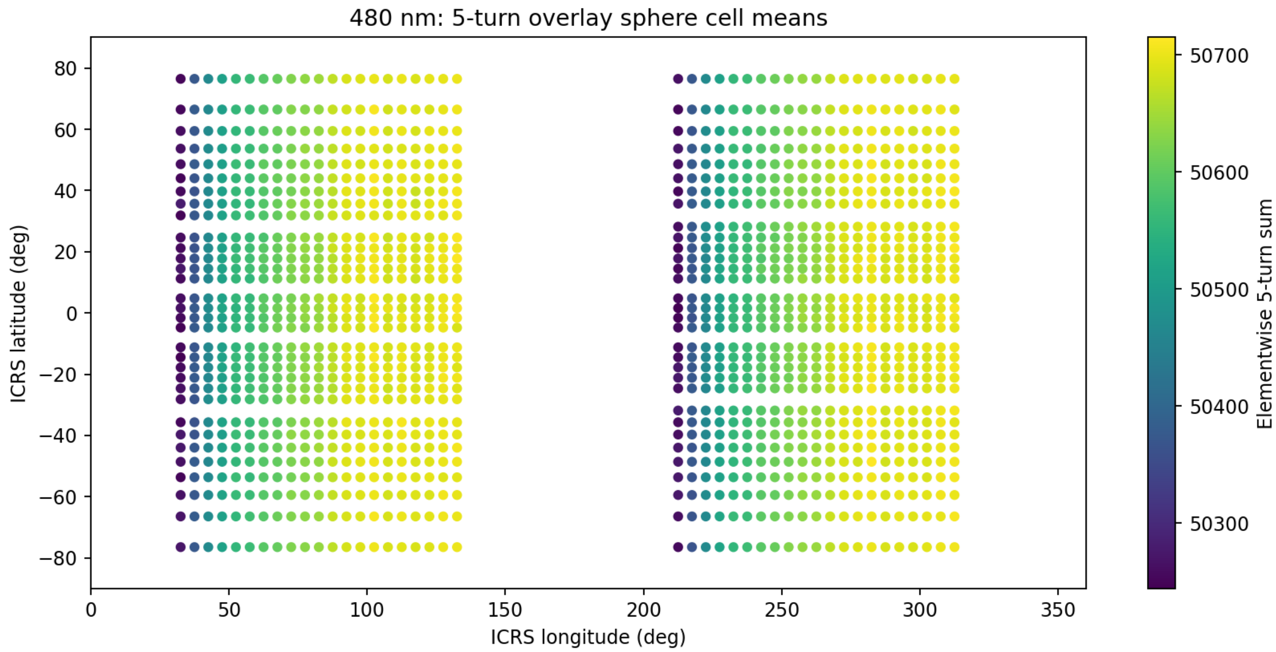

Figure 9. 480 nm five-turn surface cell means. This figure is generated directly from the 515 overlapping five-turn profiles and their center-registered ICRS coordinates.

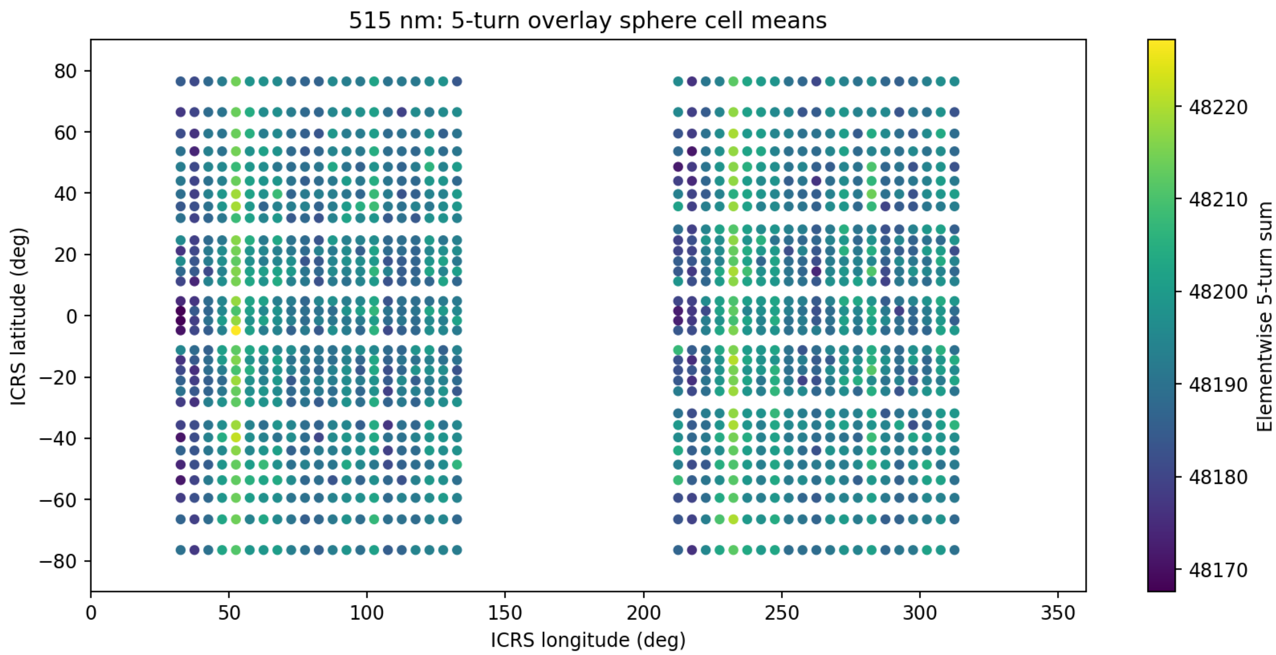

Figure 10. 515 nm five-turn surface cell means. This figure is generated directly from the 515 overlapping five-turn profiles and their center-registered ICRS coordinates.

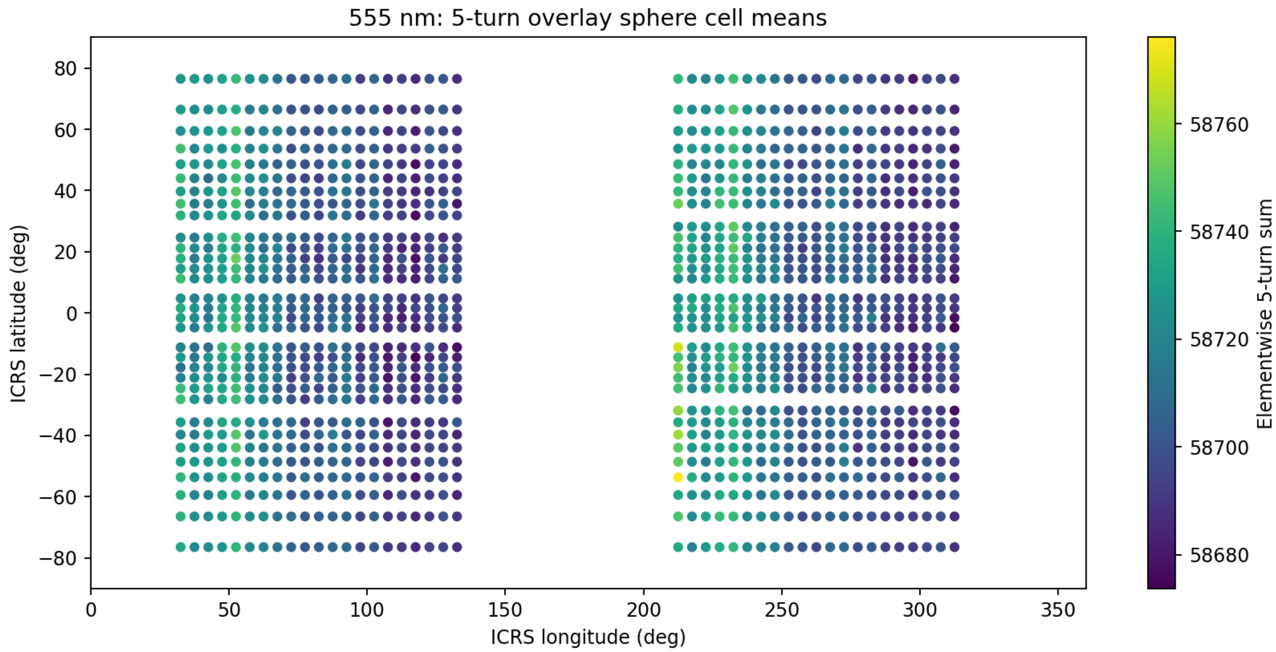

Figure 11. 555 nm five-turn surface cell means. This figure is generated directly from the 515 overlapping five-turn profiles and their center-registered ICRS coordinates.

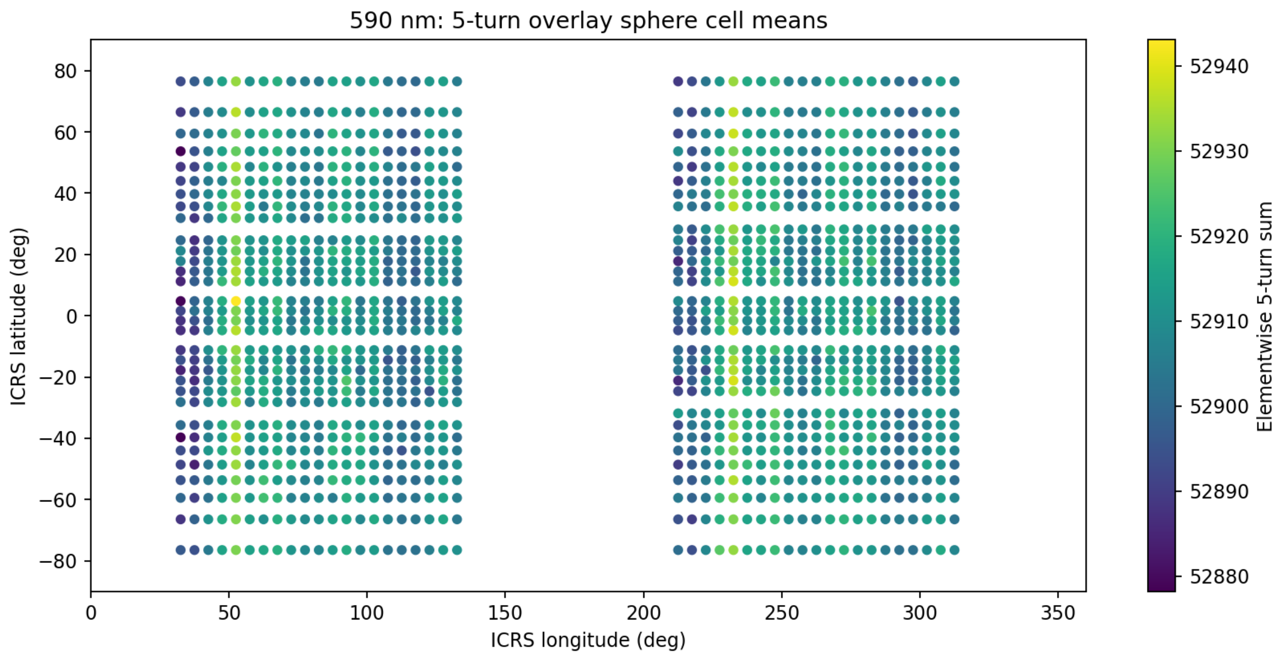

Figure 12. 590 nm five-turn surface cell means. This figure is generated directly from the 515 overlapping five-turn profiles and their center-registered ICRS coordinates.

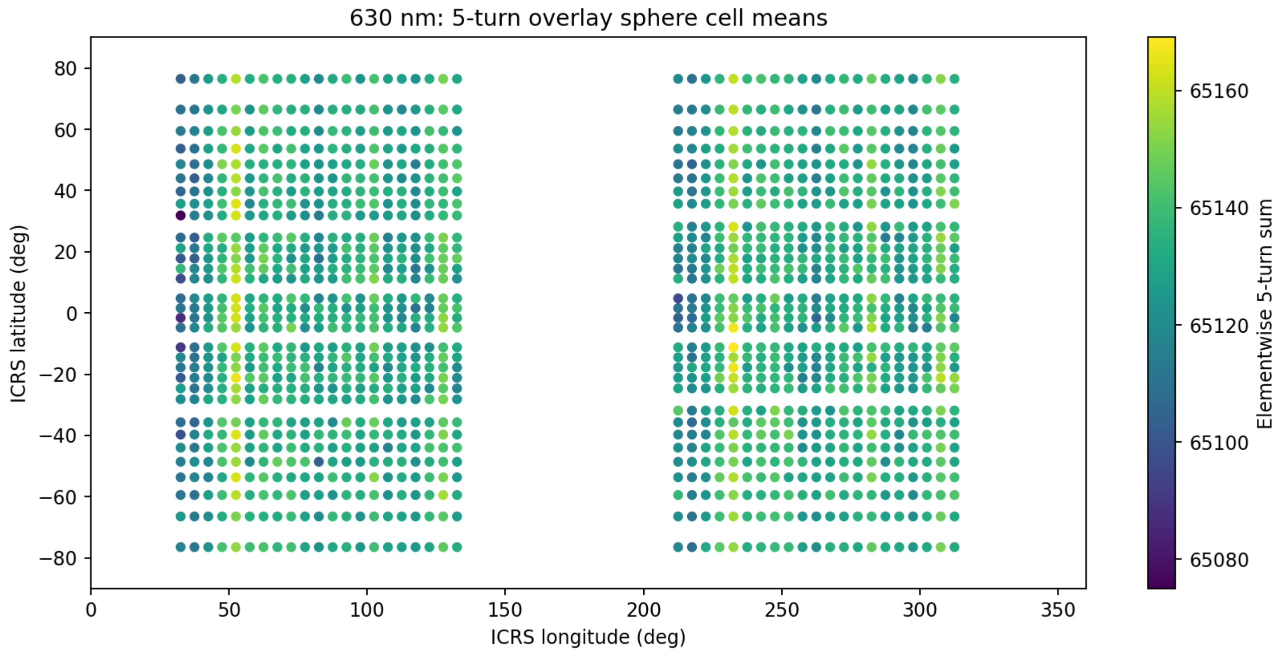

Figure 13. 630 nm five-turn surface cell means. This figure is generated directly from the 515 overlapping five-turn profiles and their center-registered ICRS coordinates.

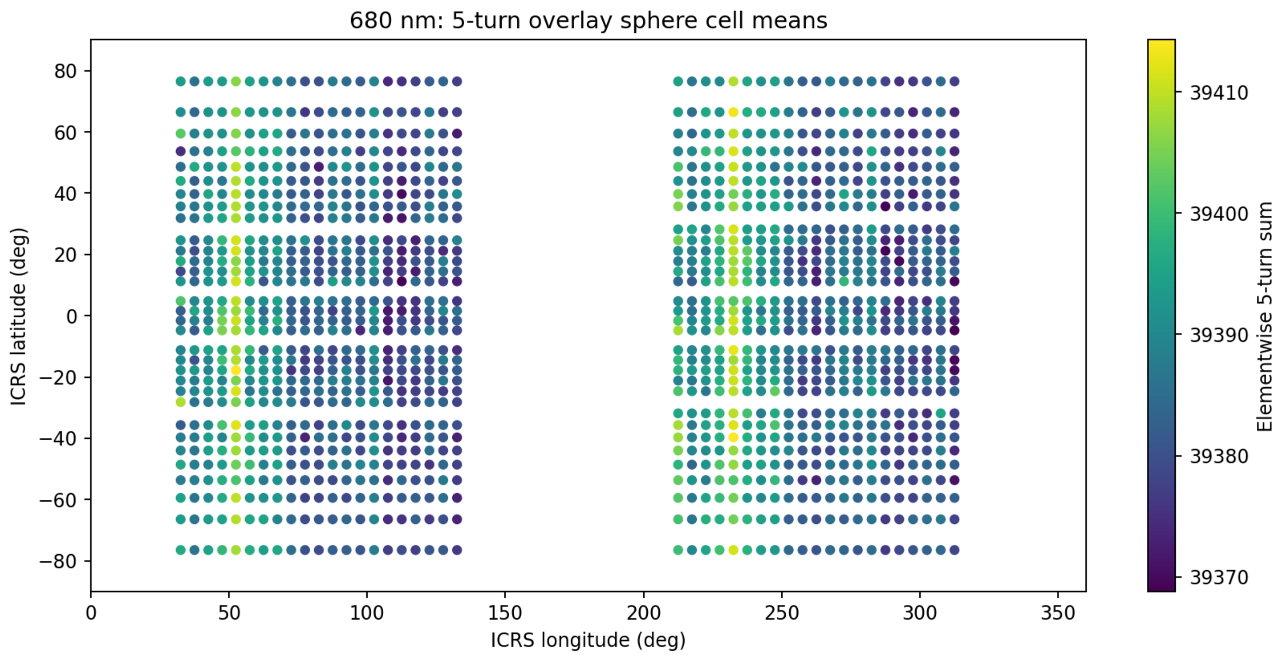

Figure 14. 680 nm five-turn surface cell means. This figure is generated directly from the 515 overlapping five-turn profiles and their center-registered ICRS coordinates.

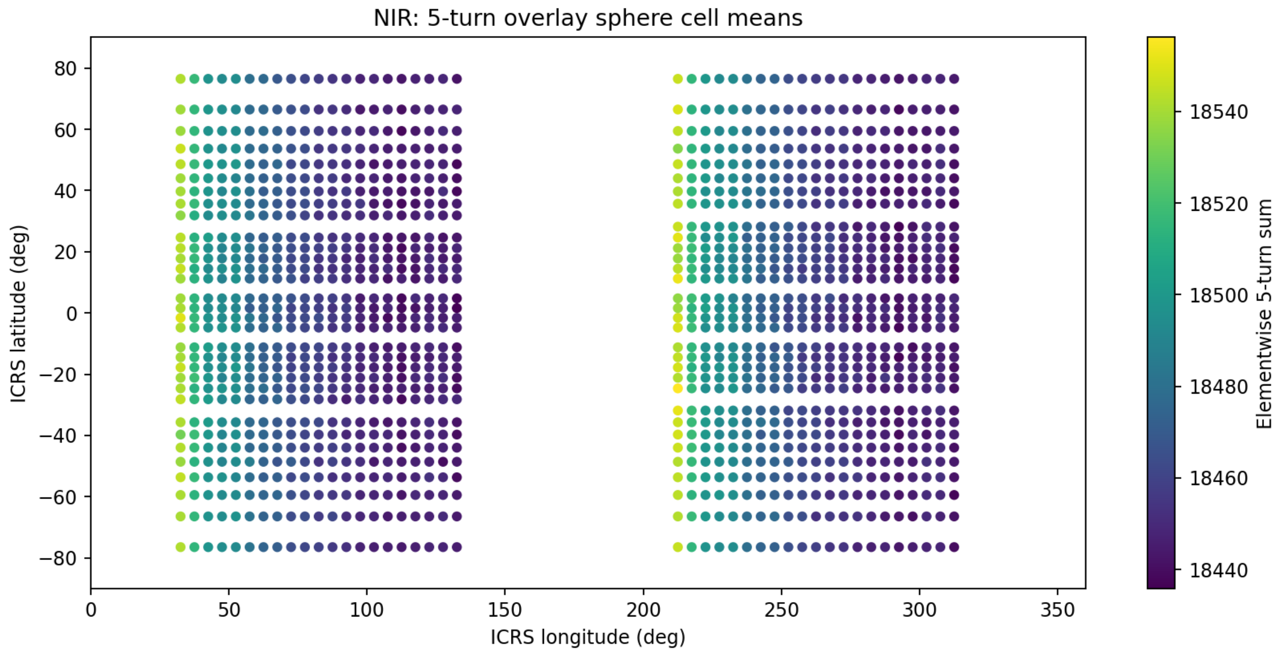

Figure 15. NIR five-turn surface cell means. This figure is generated directly from the 515 overlapping five-turn profiles and their center-registered ICRS coordinates.

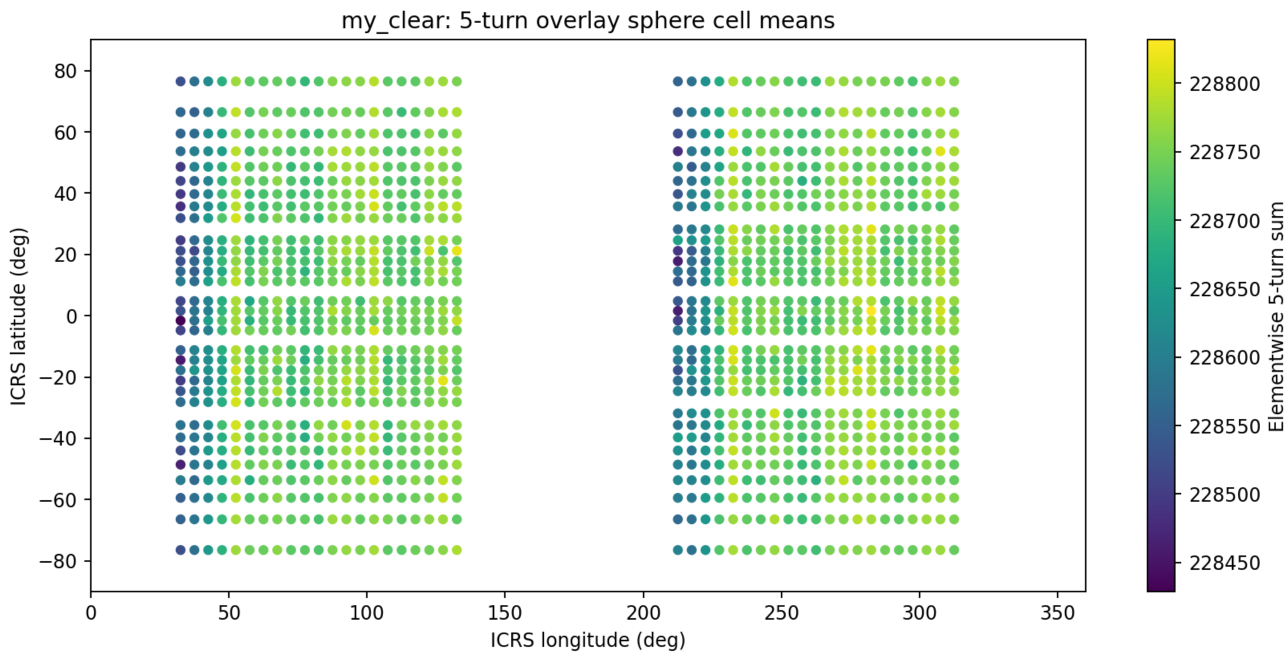

Figure 16. Clear five-turn surface cell means. This figure is generated directly from the 515 overlapping five-turn profiles and their center-registered ICRS coordinates.

9. Degree-2 Geometry and Principal Planes

9.1 Dominant harmonic degree

Every channel has dominant spherical-harmonic degree $l = 2$. The five-turn even-degree power fraction ranges from 0.890896 for 680 nm to 0.999862 for 480 nm. This identifies a common broad quadrupolar and strongly antipodal topology. The five-turn reconstruction changes coherence strength but does not change this shared degree classification.

9.2 Trace-free quadratic representation

A real degree-2 spherical field can be written as a trace-free quadratic form on the unit sphere. For each channel, the occupied cell means were weighted by sample count, standardized to zero mean and unit weighted standard deviation, and fitted as

```
q(u) = u^T Q u, with Q = Q^T and trace(Q) = 0
```

The three orthonormal eigenvectors of Q define the maximum, intermediate, and minimum quadrupole axes. Each axis also defines a principal great-circle plane through the origin by n dot u = 0. These planes are dimensionless orientation objects in the common ICRS unit-sphere coordinate system.

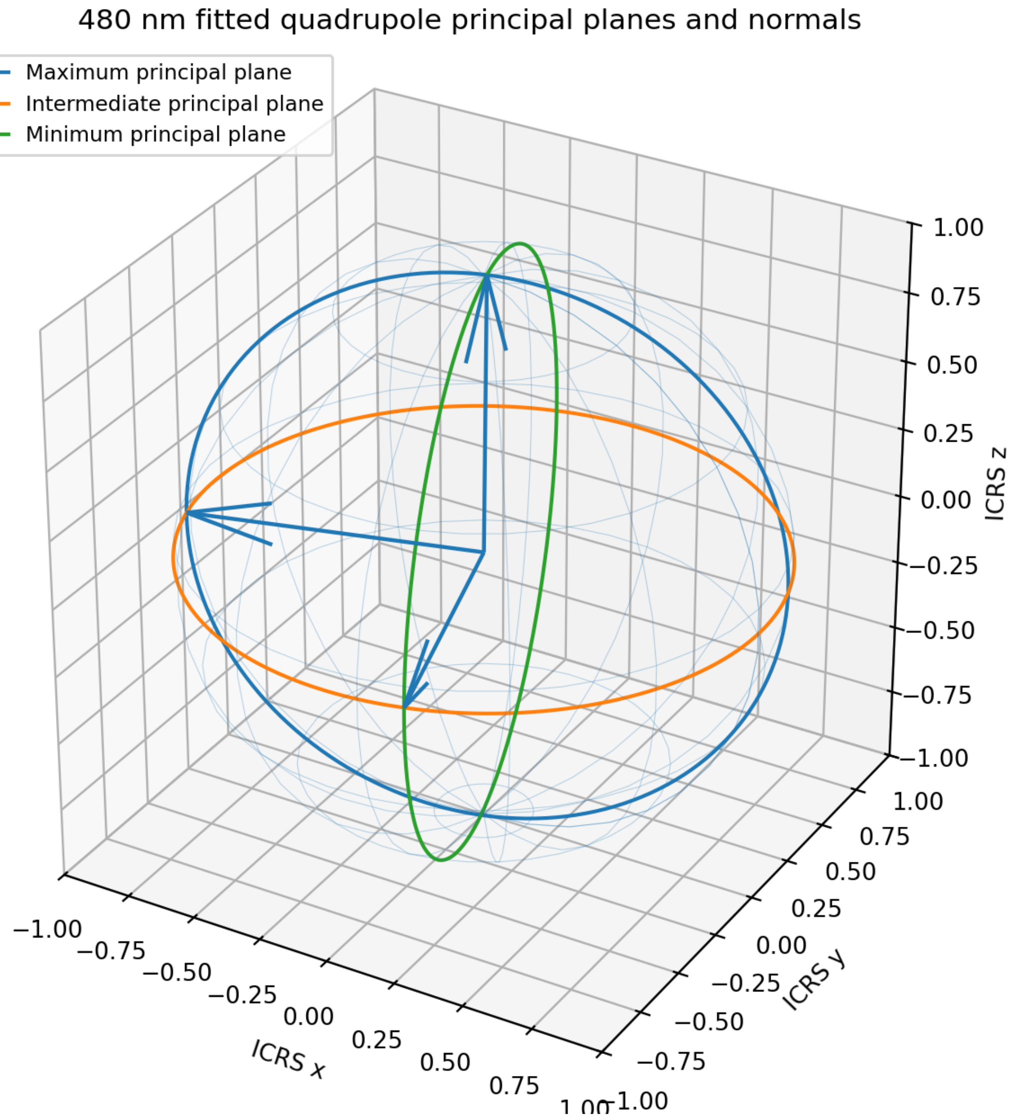

Figure 17. Dimensionally correct principal-plane geometry for the measured 480 nm five-turn map. The unit sphere has radius one. The three great circles are planes through the common origin, and the plotted normals are the fitted orthonormal eigenvectors of the channel degree-2 tensor.

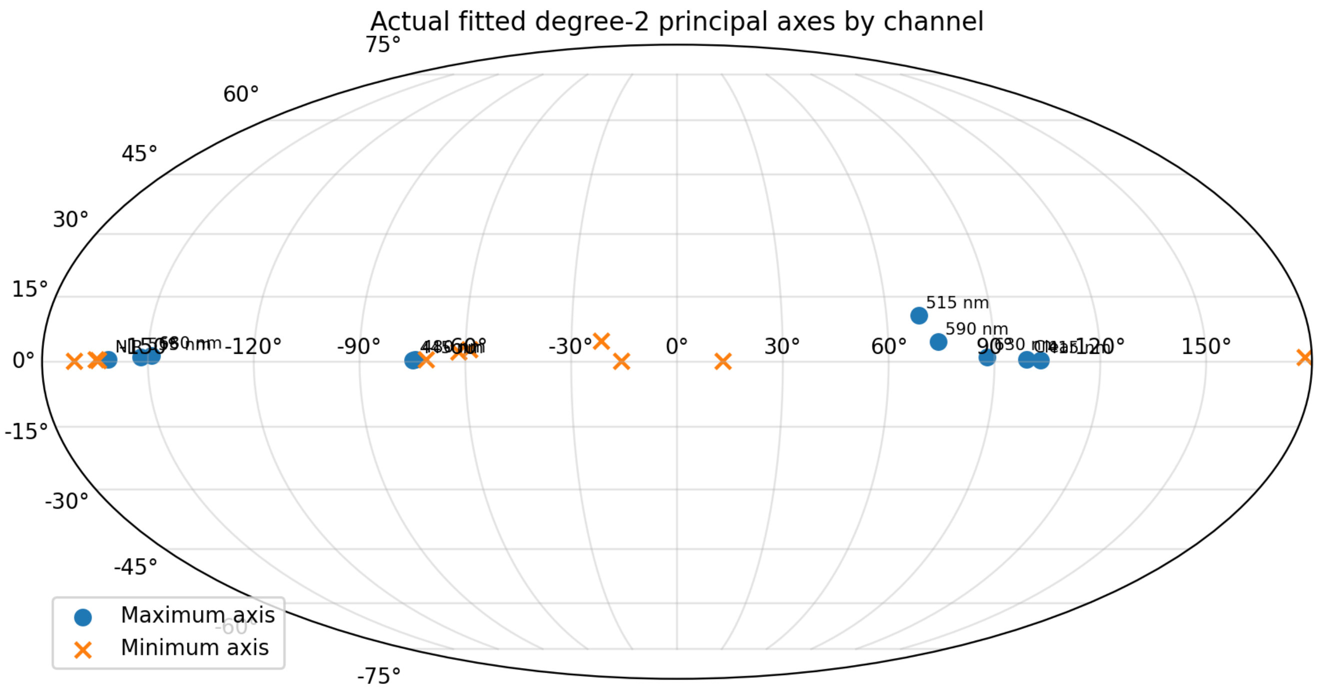

Figure 18. Actual maximum and minimum quadrupole axes for all ten channels in ICRS right ascension and declination. Labels mark the maximum-axis positions. Axis directions are projective: reversing an axis gives the same principal plane.

Channel	Quadratic R^2	Max RA	Max Dec	Mid RA	Mid Dec	Min RA	Min Dec
415 nm	0.5371	102.951	0.385	253.109	89.556	12.949	0.221
445 nm	0.5824	285.131	0.354	48.260	89.351	195.128	0.543
480 nm	0.6138	285.790	0.500	64.758	89.338	195.787	0.435
515 nm	0.0171	69.301	10.778	224.389	78.146	338.370	4.876
555 nm	0.5115	207.879	1.064	92.653	87.505	297.921	2.257
590 nm	0.0807	74.133	4.600	253.184	85.399	344.127	0.076
630 nm	0.0079	87.940	0.994	315.216	88.534	177.958	1.076
680 nm	0.3183	211.065	1.370	95.305	86.850	301.133	2.836
NIR	0.6345	198.734	0.525	62.579	89.271	288.739	0.505
Clear	0.3164	99.014	0.566	289.845	89.424	189.015	0.108

The fitted axes provide an explicit plane ledger for the degree-2 component. The 445 nm and 480 nm maximum axes are nearly coincident near RA 285 deg and the equatorial plane. Their minimum axes are likewise near RA 195 deg. NIR has a distinct equatorial maximum near RA 199 deg. The intermediate axes of the high-coherence channels lie close to the ICRS polar direction. These relationships are numerical properties of the measured five-turn maps and are preserved in the companion JSON as full Cartesian vectors, eigenvalues, tensors, and plane equations.

10. Interpretation of the Findings

10.1 Direct observations

The following statements are direct properties of the measured record and analysis: the source contains 519 complete 90-position revolutions; five-turn summation produces 515 overlapping profiles; the center-registered trajectory covers 1,344 equal-area sphere cells; eta^2 increases from three turns to five turns in every channel; the 480 nm, 445 nm, and NIR maps have the highest five-turn coherence; every channel is dominated by degree l = 2; and every channel retains strong even-degree power.

10.2 Statistical structure

The systematic increase across all channels is the central statistical result. A random channel-specific fluctuation would not be expected to produce a uniform positive five-minus-three eta^2 change across ten simultaneous channels while preserving the same dominant spherical degree. The result instead shows that the signal representation is integration-length dependent: additional phase-aligned revolutions improve the recoverable spatial state.

10.3 K-Field analysis paradigm

Within the K-Field experimental framework, the detector channels are probe responses sampled while the apparatus rotates and Earth carries the laboratory through a changing inertial orientation. The relevant observable is therefore not a single closed laboratory circle. It is a multi-turn timewise signal embedded in the translating and rotating inertial trajectory. The five-turn result establishes the practical analysis rule: resolve the channel state first in device phase, then map the resolved state to the common sphere.

10.4 SNR and angular-resolution trade

The five-turn operation increases coherent integration while introducing less than 0.769 deg of full first-to-last Earth-rotation smear. This is below the 4 deg device-index interval and below the approximately 3.989 deg equivalent cell scale. For this record, the SNR gain therefore dominates the inertial angular-smearing cost. Longer windows will continue to improve noise rejection only until signal evolution and angular smear begin to reduce coherence.

10.5 Null interpretation under overlap

The split-map and circular-shift statistics must be interpreted with the overlap structure. Consecutive five-turn profiles share 80 percent of their source revolutions. A one-turn shift therefore does not create an independent surrogate. It changes the ICRS registry while retaining extensive source content. This explains why some direct eta^2 values are at the top of the null family while their split-null percentiles remain moderate. The direct coherence, overlap-length progression, and harmonic geometry together are more informative than treating 515 windows as 515 independent trials.

10.6 What the five-turn result adds

The single-turn maps showed broad structure but could not isolate the signal-resolution timescale from the noise floor. The three-turn analysis demonstrated coherent recovery. The five-turn analysis extends that result: every channel improves again, the topology remains degree 2, and the weaker channels show the largest gains. The observed progression establishes a measured coherence-growth curve rather than a one-window result.

11. Educational Reconstruction Procedure

The complete procedure can be reconstructed without hidden steps.

Step	Operation
1	Read the source in acquisition order and identify complete index sequences 0 through 89.
2	Construct a revolution-by-position-by-channel array with shape 519 by 90 by 10.
3	For every start revolution k = 0 through 514, sum revolutions k through k + 4 elementwise.
4	Assign each resulting 90-position profile to the effective sample times of center revolution k + 2.
5	At every center time, compute the ICRS local north and up basis at latitude 42.129 deg north and longitude 83.144 deg west.
6	Map each device index to u = sin(theta) north - cos(theta) up with theta = 4 deg times index.
7	Bin u into 72 longitude bins and 36 equal-sin(latitude) bands.
8	Compute per-cell counts and means for each channel.
9	Compute sphere eta^2, alternating-block split correlation, 64 whole-turn circular-shift null values, and a real harmonic fit through l = 12.
10	Compare the results to single-turn and three-turn representations using the same metrics and grid.

11.1 Reference pseudocode

```
for k in range(0, 519 - 5 + 1):
stacked[k, :, c] = sum(turn[k+j, :, c] for j in range(5))
time[k, :] = effective_time[k+2, :]
direction[k, i] = sin(4*i deg)*north_ICRS(time[k,i]) - cos(4*i deg)*up_ICRS(time[k,i])
```

11.2 Reproducibility requirements

A valid reproduction must preserve acquisition order, complete-turn boundaries, device index, effective timing, center registration, one-revolution cadence, and elementwise summation. Averaging whole revolutions into a single scalar, randomly reordering samples, subtracting a trend on the three-to-ten-turn signal scale, or mapping each term of a five-turn stack to five separate sphere locations would implement a different analysis.

12. Conclusions

The five-revolution elementwise overlay was completed with one-revolution cadence. Each result was registered to the center revolution. The 519 complete revolutions produced 515 overlapping five-turn stacks, or 46,350 sphere-mapped samples per channel. No detrending was applied.

Relative to the three-turn overlay, sphere coherence increased in every channel. The largest gains occurred in the weaker channels: 415 nm increased from 0.656 to 0.757; 515 nm from 0.245 to 0.349; 555 nm from 0.479 to 0.600; 590 nm from 0.298 to 0.407; 630 nm from 0.224 to 0.317; 680 nm from 0.322 to 0.439; and Clear from 0.465 to 0.585. The strongest channels also improved: 480 nm reached eta^2 = 0.970 with split-map correlation 0.993, while 445 nm and NIR reached eta^2 = 0.934 and 0.909.

Every channel remains dominated by spherical-harmonic degree $l = 2$, with strong even-degree power. The five-turn integration therefore increases SNR without changing the underlying broad quadrupolar and antipodal surface geometry.

The improvement from three to five revolutions is systematic. The coherent sphere structure continues to resolve as additional consecutive revolutions are combined. For the measured revolution period, angular smearing from Earth rotation remains smaller than both the device angular sampling interval and the equal-area grid scale. The five-turn window therefore provides a favorable point on the measured SNR-resolution trade curve.

Foundational result: the correct observable for this experiment is a phase-aligned multi-revolution state embedded in inertial coordinates. Single-turn samples are components of that state, not its fully resolved representation. This result provides the analytical foundation for future overlap-length optimization, channel-coupling comparison, principal-plane tracking, and prediction of recurrent inertial crossings.

Appendix A. Complete One-, Three-, and Five-Turn Metric Ledger

Channel	eta 1	eta 3	eta 5	split 1	split 3	split 5	R2 1	R2 3	R2 5
415 nm	0.396506	0.656222	0.757237	0.885326	0.900259	0.912163	0.860112	0.906345	0.907301
445 nm	0.788082	0.904446	0.933965	0.971639	0.974766	0.977260	0.915218	0.950892	0.951382
480 nm	0.920165	0.960656	0.969747	0.989930	0.991469	0.993206	0.934876	0.966784	0.968012
515 nm	0.101413	0.245332	0.349337	0.531914	0.566049	0.598008	0.323535	0.421596	0.426777
555 nm	0.247466	0.478892	0.600390	0.792946	0.811744	0.833279	0.783931	0.812244	0.814444
590 nm	0.133405	0.297836	0.407008	0.577401	0.591198	0.612596	0.391608	0.484044	0.487020
630 nm	0.094315	0.224225	0.316982	0.490476	0.491013	0.504538	0.284544	0.377194	0.378243
680 nm	0.145318	0.322415	0.438892	0.644775	0.682899	0.718098	0.605168	0.659620	0.664832
NIR	0.700506	0.865384	0.909296	0.969399	0.973906	0.977249	0.936814	0.960474	0.962122
Clear	0.239664	0.464965	0.585296	0.740388	0.763241	0.789715	0.683567	0.747384	0.745580

Appendix B. Principal Plane Equations

Each equation below defines the great-circle plane normal to the listed fitted axis. Coordinates x, y, and z are dimensionless ICRS unit-sphere coordinates. Axis signs are conventional; multiplying all coefficients by -1 gives the identical plane.

Channel	Plane	Eigenvalue	Plane equation
415 nm	Maximum	+1.893533	-0.224104883220 x +0.974541878064 y +0.006717828204 z = 0
415 nm	Intermediate	+0.636078	-0.002250304679 x -0.007410590720 y +0.999970009187 z = 0
415 nm	Minimum	-2.529611	+0.974562433837 x +0.224083044972 y +0.003853765716 z = 0
445 nm	Maximum	+1.886029	+0.261025438578 x -0.965312098288 y +0.006186542940 z = 0
445 nm	Intermediate	+0.606439	+0.007535749224 x +0.008446148001 y +0.999935935482 z = 0
445 nm	Minimum	-2.492467	-0.965302508491 x -0.260962095873 y +0.009479009382 z = 0
480 nm	Maximum	+1.914096	+0.272107468501 x -0.962227353863 y +0.008720382089 z = 0
480 nm	Intermediate	+0.602392	+0.004929583757 x +0.010456131731 y +0.999933182024 z = 0
480 nm	Minimum	-2.516489	-0.962254241243 x -0.272046298977 y +0.007588571870 z = 0
515 nm	Maximum	+0.371399	+0.347229780400 x +0.918945565091 y +0.187004085525 z = 0
515 nm	Intermediate	+0.111057	-0.146790535713 x -0.143693150934 y +0.978675031356 z = 0
515 nm	Minimum	-0.482457	+0.926220286017 x -0.367275546115 y +0.084997970547 z = 0
555 nm	Maximum	+1.784933	-0.883786793420 x -0.467521256737 y +0.018568206023 z = 0
555 nm	Intermediate	-0.231518	-0.002015289065 x +0.043488263205 y +0.999051905345 z = 0
555 nm	Minimum	-1.553415	+0.467885501363 x -0.882911459583 y +0.039376543165 z = 0
590 nm	Maximum	+0.901800	+0.272526163106 x +0.958799391852 y +0.080207335118 z = 0
590 nm	Intermediate	+0.420140	-0.023207725478 x -0.076787846227 y +0.996777321246 z = 0
590 nm	Minimum	-1.321940	+0.961868437938 x -0.273509328645 y +0.001324855969 z = 0
630 nm	Maximum	+0.317558	+0.035948313716 x +0.999202925934 y +0.017356023359 z = 0
630 nm	Intermediate	+0.138139	+0.018153505434 x -0.018017294539 y +0.999672860159 z = 0
630 nm	Minimum	-0.455698	-0.999188755432 x +0.035621480926 y +0.018786727102 z = 0
680 nm	Maximum	+1.359474	-0.856333160602 x -0.515869818846 y +0.023913344768 z = 0
680 nm	Intermediate	-0.156808	-0.005081061941 x +0.054719792169 y +0.998488821747 z = 0
680 nm	Minimum	-1.202666	+0.516398780850 x -0.854917583367 y +0.049479539079 z = 0
NIR	Maximum	+2.376988	-0.946978446659 x -0.321166167757 y +0.009171382238 z = 0
NIR	Intermediate	-0.511160	+0.005856054488 x +0.011287251599 y +0.999919149020 z = 0
NIR	Minimum	-1.865828	+0.321243720856 x -0.946955590638 y +0.008808017408 z = 0

Channel	Plane	Eigenvalue	Plane equation
Clear	Maximum	+1.570813	-0.156665077762 x +0.987602398515 y +0.009877036883 z = 0
Clear	Intermediate	+0.611928	+0.003413802354 x -0.009458982541 y +0.999949435523 z = 0
Clear	Minimum	-2.182742	-0.987645887636 x -0.156690874326 y +0.001889587157 z = 0

Appendix C. Source and Output Data Dictionary

Field	Definition
record_id	Source row identifier.
a415nm through a680nm	Native detector counts for nominal spectral channels.
NIR	Near-infrared channel count.
my_clear	Broadband Clear channel count.
my_index	Mechanical position index 0 through 89; angular step is 4 deg.
frame_read_us	Frame-read duration in microseconds, subtracted from date_UTC for effective time.
date_UTC	Recorded UTC timestamp.
stack_start_turn	First source revolution in a five-turn window.
stack_center_turn	Center source revolution used for ICRS registration.
stack_end_turn	Fifth source revolution in the window.
ICRS_x, ICRS_y, ICRS_z	Dimensionless center-registered direction vector components.
sphere_cell	Integer equal-area cell identifier from 0 through 2591.
cell_mean	Mean five-turn elementwise sum for all mapped samples in a cell.
harmonic_fit	Degree-0 through degree-12 regularized real spherical-harmonic prediction at the cell center.

Appendix D. Machine-Readable Companion

The companion file K-Field_spectrophtometer_spheroical_coherence_1785873414.json contains report metadata, source hashes, exact timing values, the complete grid-cell definition, all one-turn, three-turn, and five-turn channel results, all fitted degree-2 tensors and plane equations, and all 13,440 occupied channel-cell records from the five-turn surface maps. The full 46,350-row center-registered coordinate ledger remains in pic_id14A_5turn_overlay_unit_sphere_coordinates.csv and is referenced by filename and hash.

Artifact	Purpose
K-Field_spectrophtometer_spheroical_coherence_1785873414.json	Archival machine-readable companion.
K-Field_spectrophtometer_spheroical_coherence_1785873414.pdf	Human-readable archival reference.
pic_id14A_5turn_overlay_unit_sphere_coherence_summary.csv	Five-turn channel summary.
pic_id14A_5turn_overlay_unit_sphere_surface_maps.csv	Five-turn occupied-cell means and harmonic values.
pic_id14A_5turn_overlay_unit_sphere_coordinates.csv	Five-turn center-registered sample coordinates and stacked channel values.
pic_id14A.csv	Original source record.

Appendix E. Reproduction Record

Field	Value
Principal Investigator	P. Dwayne Esterline, Principal Investigator
Degrees	BS, Huntington University (1999); MBA, Indiana Wesleyan University (2008)
Organization	Krytronx, LLC.
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.
Generation epoch	1785873414
Generation UTC	2026-08-04T19:56:54Z
PDF	K-Field_spectrophtometer_spheroical_coherence_1785873414.pdf
JSON	K-Field_spectrophtometer_spheroical_coherence_1785873414.json
PDF engine	Python ReportLab
PDF fonts	Helvetica, Helvetica-Bold, Times-Roman, Courier only
Source record	pic_id14A.csv
Source SHA-256	f86bff218ebf3d20debdc651d87f5bb27fe74beebaa2964ea505658189f9caf1
Five-turn summary SHA-256	539bdfa8610dff66ca5269b7ae9af6c5d8cd9fbab6e69fb3a4260dff802a3f78
Five-turn surface-map SHA-256	76c1c1ab7ac45caf370f789b659a26752fe81c4a81be26bb3d9579e457acb633
Five-turn coordinate SHA-256	dc08da7097a8f6d24aceb0b3b0dfb61384382412c256850b097ef4a59b1f9556
Historical format references	K-Field Spectrophotometer Crossing Analysis; K-Field Spectrophotometer Optimized Geometry; K-Field 7x15 Spectrophotometer Interleaved Analysis

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